



ZIMBABWE

**MINISTRY OF HIGHER AND TERTIARY EDUCATION,
SCIENCE AND TECHNOLOGY DEVELOPMENT**

**HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)**

NATIONAL CERTIFICATE

IN

**MOTOR MECHANICS, AUTO ELECTRICS AND ELECTRONICS MVBR, DPF,
MOTOR CYCLE MECHANICS AND PRECISION MACHINING**

SUBJECT: Engineering Mathematics PAPER NO: 346/13/S08

DURATION: 3 Hours

OCTOBER/NOVEMBER 2017 EXAMINATION

REQUIREMENTS

Scientific Calculator and Graph Paper

INSTRUCTIONS TO CANDIDATE

Section A Answer all questions (40 marks)

**Section B Choose only five questions each question carries 12 marks
total (60 marks)**

This paper consists of 4 printed pages.

FOOD₁₆

SECTION A
QUESTION 1

- a) Convert $FOOD_{16}$ to denary (5 marks)
- b) By the quadratic formula solve:
 $2x^2 + 5x = 3$ (3 marks)
- c) Simplify $\frac{2a-3}{4b} + \frac{3a+2}{6b}$ (4 marks)

QUESTION 2

- a) Find the remainder if $3x^3 - 2x^2 + x - 5$ is divided by $x - 1$ using remainder theorem. (2 marks)
- b) Solve the cubic function $x^3 - 2x^2 - x + 2$ using factor theorem (3 marks)
- c) Simplify $F = \sqrt[3]{a^6 b^3} \div \sqrt{\frac{1}{9} a^4 b^6} \times (4\sqrt{a^6 b^2})^{\frac{1}{2}}$ (4 marks)

QUESTION 3

- a) Evaluate $\log_{10} x = 5$ (2 marks)
- b) A trapezium has parallel sides of 5cm and 9cm and a height 12cm. Find its area (3 marks)
- c) Differentiate $\sqrt{x}$ (2 marks)
- d) A straight line passing through point $A(-4; 5)$ and $B(1; 3)$ Find
i) the midpoint of the line (1 mark)
ii) the gradient of the straight line (2 marks)

QUESTION 4

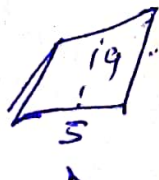
- a) Matrix $A = \begin{bmatrix} 7 & 3 \\ 4 & 8 \end{bmatrix}$ find

- i) $\frac{1}{2}A$
ii) A^{-1}

- b) Find $\int 3\sqrt{x} dx$

- c) One side of a right angled triangle is 4cm, the hypotenuse is 5cm. Calculate the missing side. (3 marks)

A B C D E F G H I J K L M N O P Q
10 11 12 13 14 15 16 17 18 19 20 21 22 23 24



6 1 4 5 $\frac{2}{3}$
10

22(256) 22(256)
5632 352
FO 3x2
x² - 3x - 2

$$\begin{array}{r} 2x^2 + x + 2 \\ x-1 \overline{) 2x^3 - 2x^2 + x - 5} \\ \underline{2x^3 - 2x^2} \\ 3x - 3x^2 \\ \underline{3x^2 - 3x} \\ 6x - 3x^2 \\ \underline{6x - 6} \\ -3x^2 + 3x - 5 \end{array}$$

$$\begin{array}{r} x^2 - x \\ x-1 \overline{) x^2 - x} \\ \underline{x^2 - x} \\ 0 \end{array}$$

$$\begin{array}{r} 3x^2 + x + 2 + 3 \\ x-1 \overline{) 3x^2 + x + 5} \\ \underline{3x^2 - 3x} \\ 4x + 5 \end{array}$$

$$\begin{array}{r} 2x - 2 \\ 2x \overline{) 2x - 2} \\ \underline{2x - 2} \\ 0 \end{array}$$

$$\begin{array}{r} 3x^2 \\ 3x \overline{) 3x^2} \\ \underline{3x^2} \\ 0 \end{array}$$

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SECTION B

QUESTION 5

Determine the volume and total surface area of the following solids.

$\frac{1}{3}\pi r^2 h$
 $\pi r L + \pi r^2$

- a) A cone of radius 8.0cm and perpendicular height 10 cm (4 marks)
b) A sphere of diameter 7.0cm (4 marks)
c) A hemisphere of radius 3.0cm (4 marks)

QUESTION 6

- a) When a flywheel 410mm diameter is rotated, it is found that a particular valve opens when the datum mark has travelled a distance of 400mm past TDC measured around the circumference. The same valve closes when the flywheel mark has moved 37mm past TDC. Calculate the opening and closing positions in degrees of crank movement and state which value is it. (8 marks)
b) A taper of 2° included angle is required on a shaft. The diameter of the small end being 13.75cm. Calculate length of shaft necessary to give a large end diameter of 14.37cm. (4 marks)

QUESTION 7

For the curve $y = 2x^2 - 7x + 3$ determine

- a) Where the graph crosses the y and x axis (4 marks)
b) The coordinates of stationary points (4 marks)
c) The nature of stationery points (2 marks)
d) Sketch the curve (2 marks)

QUESTION 8

- a) Difference $y = 7x^2 + 4x + 5$ from first principles (6 marks)
b) The distance x in meters travelled by a vehicle in time (t) seconds after the brakes are applied is given by $x = 20t - \frac{5}{3}t^2$. Determine.
i) Speed of vehicle in km/hr at the instant the brakes are applied (3 marks)
ii) Distance the car travels before it stops (3 marks)

QUESTION 9

- a) When $y = [5 + 4x - x^2]$ find $\int_1^4 (y) dx$ (6 marks)
b) A curve has a gradient function of $(x + 4)$ find the equation of the curve given that it passes through point $[2; 3]$ (6 marks)

QUESTION 10

Find the inverse of the following matrix $\begin{bmatrix} 1 & 5 & -2 \\ 3 & -1 & 4 \\ -3 & 6 & -7 \end{bmatrix}$

$20t - \frac{5}{3}t^2$

$\frac{5}{3} \times 2 = \frac{10}{3}$

$1/2$

$y = mx + c$

x_1

$y - y_1 = (x - x_1)(y_1 - x_1)$

$y = mx + c$

(12 marks)

x^2

$\frac{1}{x^2}$

$\frac{1}{x}$

$\frac{1}{2} - 1$

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QUESTION 11

Prove without using logarithms

$$\frac{\log \sqrt{27} + \log \sqrt{8} - \log \sqrt{125}}{\log 6 - \log 5} = \frac{3}{2}$$

(12)

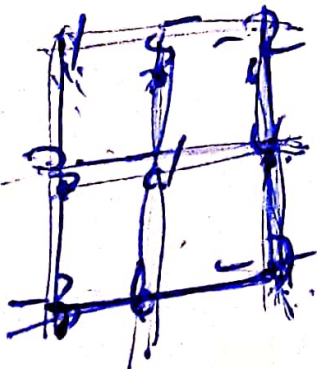
...../km

$$\begin{array}{r} x^3 - 2x^2 - x + 2 \\ (x^2 - 2x - 1)x + 2 \\ (x - 2)(x - 1)x + 2 \\ \underline{2 - 2} \end{array}$$

$$\begin{array}{r} 0 - 1 - 1 \end{array}$$

$$\begin{array}{r} 0 - 1 \\ 0 \end{array}$$

$$1 - 2$$



$$7 - 24 = -17 \checkmark$$

$$-21 - -12 = -9 \checkmark$$

$$18 - 3 = 15 \checkmark$$

$$-7 + 6$$